

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 1: Real Numbers

Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board paper and the marks. Solutions are in a separate PDF with the same question numbers.

1 HCF and LCM by prime factorisation

1. If two positive integers p and q can be expressed as $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are prime numbers, then LCM (p, q) is: (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$

CBSE 2024, 1 mark

2. In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject.

CBSE 2024, 3 marks

2 The Fundamental Theorem of Arithmetic

3. Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number. CBSE 2024, 2 marks

3 Irrational numbers and their proofs

4. Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number.

CBSE 2024, 2 marks